ARTICLE

Age-Related Crossover in Breast Cancer Incidence Rates Between Black and White Ethnic Groups

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Background Although breast cancer incidence is higher in black women than in white women among women younger than 40 years, the reverse is true among those aged 40 years or older. This crossover in incidence rates between black and white ethnic groups has been well described, has not been completely understood, and has been viewed as an artifact.

Methods To quantify this incidence rate crossover, we examined data for 440 653 women with invasive breast cancer from the National Cancer Institute's Surveillance, Epidemiology, and End Results database from January 1, 1975, through December 31, 2004. Data on invasive female breast cancers were stratified by race, age at diagnosis, year of diagnosis, and tumor characteristics. Standard descriptive analyses were supplemented with Poisson regression models, age–period–cohort models, and two-component mixture models. All statistical tests were two-sided.

Results We observed qualitative (ie, crossing or reversing) interactions between age and race. That is, age-specific incidence rates overall (expressed as number of breast cancers per 100 000 woman-years) were higher among black women (15.5) than among white women (13.1) younger than 40 years (difference = 2.4, 95% confidence interval [CI] = 2.4 to 2.4), and then, age-specific rates crossed with rates higher among white women (281.3) than among black women (239.5) aged 40 years or older (difference = 41.8, 95% CI = 41.7 to 41.9). The black-to-white incidence rate crossover was observed for all tumor characteristics assessed, although the crossover occurred at earlier ages of diagnosis for low-risk tumor characteristics than for high-risk tumor characteristics. The incidence rate crossover between ethnic groups was robust (ie, reliable and reproducible) to adjustment for calendar period and birth cohort effects in age–period–cohort models ($P < .001$ for difference by race).

Conclusion Although this ecologic study cannot determine the individual-level factors responsible for the racial crossover in vital rates, it confirms that the age-related crossover in breast cancer incidence rates between black and white ethnic groups is a robust age-specific effect that is independent of period and cohort effects.

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It is widely recognized that breast cancer incidence rates overall are higher in black women than in white women younger than 40 years. However, it is not as well known that age-specific incidence rates are higher for black women among women younger than 40 years and higher for white women among women 40 years or older. This so-called black-to-white ethnic crossover has been well described (1–4) but has not been fully understood and has sometimes been viewed as an artifact (5).

Differential screening practices and/or hormone replacement therapy use cannot explain the higher incidence rates among young black women because routine screening mammography and hormone replacement therapy begin after the age of 40 years. Differential racial profiles for age-dependent risk factors may be a more plausible explanation for the crossover (3,5–7), with black women having an elevated risk for the early-onset types of breast cancer and a reduced risk for the late-onset types of breast cancer, compared with white women. If differential risk factor profiles for different breast cancer types account for the black-to-white incidence rate crossover, then the crossover is a population-based expression of biologic and/or etiologic heterogeneity and should be a robust (ie, reliable and reproducible) feature of vital incidence rates.

To assess the reliability and reproducibility of the black-to-white breast cancer incidence rate crossover, we obtained case and

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population data from the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) database. We supplemented standard descriptive epidemiology with three mathematical models. First, Poisson regression models provided statistical significance tests of age interactions between black and white race; statistically significant age interactions would be consistent with biologic interactions due to etiologic heterogeneity (8). Second, age–period–cohort models yielded fitted age-specific incidence rate curves for black and white women that were adjusted for calendar period and/or birth cohort effects. Different age-specific incidence rate patterns after adjustment for period and/or cohort effects (or artifacts) would be consistent with different age-dependent biology and/or etiology. Third, two-component mixture models identified heterogeneity in the age distributions at diagnosis among black and white women by determining whether two or more breast cancer populations fit the data better than a single breast cancer population. Different age distributions at diagnosis in black vs white women would provide further proof of biologic heterogeneity (9,10).

Patients and Methods

We obtained data for patients with invasive female breast cancer from nine registry databases in the National Cancer Institute's SEER program (the SEER 9 Registries database) from January 1, 1975, through December 31, 2004. These nine SEER registries include Connecticut, Iowa, New Mexico, Utah, Hawaii, Detroit, San Francisco–Oakland, Atlanta, and Seattle–Puget Sound, and cover approximately 10% of the US population (11). We stratified breast cancer patients by black and white race, age at diagnosis, year of diagnosis, and estrogen receptor (ER) expression status. Other racial groups were excluded from this analysis. The following four age groups were arbitrarily chosen to approximate mammography screening prevalence and/or use of hormone replacement therapy: age younger than 40 years (no routine screening or use of hormone replacement therapy), ages 40–49 years (low screening and no use of hormone replacement therapy), ages 50–69 years (likely screened and highest use of hormone replacement therapy), and age 70 years or older (low screening and very low or no use of hormone replacement therapy).

Demographic and Tumor Characteristics

Although black and white race, age at diagnosis, and tumor grade were recorded for the entire study period in the SEER database, extent of disease codes for tumor size and axillary lymph node involvement was most consistently recorded from 1988 forward. Estrogen receptor status was collected from 1990 forward.

Tumor characteristics (tumor size, lymph node involvement, grade, and ER status) were dichotomized into favorable (low risk) vs unfavorable (high risk) groups (ie, tumor size ≤2.0 vs >2.0 cm, lymph node–negative vs lymph node–positive status, low- vs high-grade tumor, and ER-positive vs ER-negative status, respectively). Low-grade tumors included grade 1 (well differentiated) plus grade 2 (moderately differentiated). High-grade tumors included grade 3 (poorly differentiated) plus grade 4 (undifferentiated or anaplastic). Each SEER registry recorded ER status as positive, negative, missing, borderline, or unknown. For all tumor characteristics (tumor size, lymph node involvement, grade, and ER status), missing, borderline, or unknown tumor features were combined into a single group that was designated "other or unknown" for descriptive purposes but were excluded from further analyses.

CONTEXT AND CAVEATS

Prior knowledge

The incidence rate for breast cancer among women younger than 40 years is higher among black women than among white women, but the incidence rate among women aged 40 years or older is higher among white women than among black women.

Study design

Population-based retrospective analysis of data for 440 653 women with invasive breast cancer from the National Cancer Institute's Surveillance, Epidemiology, and End Results database from January 1, 1975, through December 31, 2004.

Contribution

The age-specific ethnic crossover in incidence rates for breast cancer was confirmed.

Implications

The breast cancer incidence rate crossover between black and white ethnic groups may reflect breast cancer heterogeneity, with black women having more early-onset and less late-onset types of breast cancer than white women.

Limitations

This study had the usual limitations of a descriptive epidemiologic study, including a retrospective registry assessment, missing data, nonstandardized histopathologic typing, and lack of individual-level risk factor data.

From the Editors

Statistical Methods

Age-standardized (US standard population for the year 2000) incidence rates were obtained by use of SEER*Stat 6.3.6 statistical software and expressed as the number of breast cancers per 100 000 woman-years (ie, women per year). Percent changes in incidence rates were calculated from the earliest to the latest recorded time periods; 95% confidence intervals for the percent changes were calculated by the delta method (12).

Age-standardized incidence rates were graphed on log–linear plots by six 5-year time periods (1975–1979, 1980–1984, 1985–1989, 1990–1994, 1995–1999, and 2000–2004). Age-specific incidence rates were graphed on a log–log scale by fourteen 5-year age groups (ages 20–24, 25–29, …, 80–84, and ≥85 years). Poisson regression models were used to examine the association of age and race with breast cancer incidence rates. For this analysis, we used cubic regression splines to obtain a smooth but flexible model of the age effects. We then selected a common value for the number of segments by use of the Akaike information criterion (13) and tested for an interaction between age (modeled as a regression spline) and race by use of a likelihood ratio test. Under the null hypothesis of no interaction between age and race, the age-specific incidence rate curves for black and white women would be parallel on the log–log

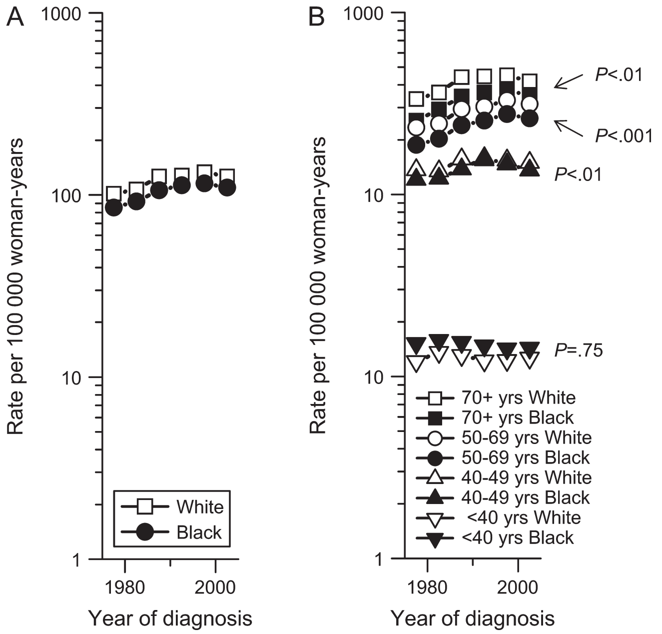

Figure 1. Age-standardized (US population in the year 2000) temporal trends and age-specific incidence trends for breast cancer among white and black women in the National Cancer Institute's Surveillance, Epidemiology, and End Results 9 Registries database from 1975 through 2004. **A**) Overall age-standardized incidence rate trends by race. **B**) Age-specific incidence rate trends by race (black vs white) and age at diagnosis (age groups ≥70, 50–69, 40–49, and <40 years). Poisson regression models were used to obtain *P* values to assess age interactions by race within age groups. All statistical tests were two-sided.

scale or proportional on the absolute scale. Statistically significant age interactions could be quantitative (noncrossover) or qualitative (8,14,15). Quantitative interactions correspond to slope changes in incidence rates in magnitude but not in direction (eg, the black vs white incidence rate ratio [IRR_{BW}] was always >1.0 or always <1.0). Qualitative interactions correspond to slope changes in magnitude and direction with the age lines crossing (eg, the black-to-white incidence rate ratio switched from >1.0 to <1.0 or from <1.0 to >1.0). We also compared age-specific incidence rates by ER expression status with Poisson regression models (Supplementary Figures 1 and 2, available online). As for race, statistically significant interactions between age ER status could be quantitative (ie, noncrossover) or qualitative. Even though ER status is a tumor characteristic rather than an independent predictor variable, parameterization of ER status allowed us to test for different age-related effects by ER expression by use of a likelihood ratio test.

Age–period–cohort models were used to simultaneously assess age-specific incidence rates that are adjusted for calendar period and birth cohort effects (16–19). As described by Tarone and Chu (20), we used 2-year age groups and 2-year time periods. Given that the year of birth equals the year of diagnosis minus the age at diagnosis, our analysis of fifteen 2-year time periods from 1975 through 2004 and thirty-two 2-year age groups from ages 20 through 83 years spanned forty-six 2-year birth cohorts from 1893 to 1983 (referred to by midyear of birth). Because the age, period, and cohort variables are collinear, it is not possible to completely separate the linear trend in calendar period effects from the linear trend in age effects or the linear trend in birth cohort effects from the linear trend in calendar period effects (18). This collinearity is known as the "nonidentifiability" problem of age–period–cohort models. Notwithstanding this nonidentifiability issue, certain age–period–cohort parameters can be estimated if the age, calendar period, and birth cohort trends are orthogonally decomposed into their linear and nonlinear components (19) (for additional details, see Appendix 1).

In brief, the estimable age–period–cohort parameters include intercepts, "drifts" (or linear trends) (16), "deviations" (or nonlinear departures from the linear trends) (21), "curvatures" (or second differences) (17), and "slope contrasts" (or differences between log–-linear trends over adjacent blocks of ages, periods, or cohorts) (22). Another useful identifiable age–period–cohort parameter is the "fitted age of onset curve," which is the sum of an intercept term, a drift parameter that we refer to as the "longitudinal age trend" (16,23), and an age deviation (18). We fitted the age-at-onset curves to breast cancer incidence for white and black women separately. Because the curves were obtained from separate age–period–cohort models, they also were adjusted for any period and

Figure 2. Age-specific incidence rates for breast cancer among white and black women in the National Cancer Institute's Surveillance, Epidemiology, and End Results 9 Registries database from 1975 through 2004. **A**) Age-specific incidence rates. Age-specific incidence rates were higher among black women than among white women younger than 40 years and higher among white women aged 40 years or older. This so-called incidence rate crossover between black and white ethnic groups was near Clemmesen's menopausal hook, which occurs at approximately age 50 years. Poisson regression models were used to obtain *P* values to assess age interactions by race. All statistical tests were two-sided. **B**) Age-specific incidence rate curves from the age–period–cohort fitted model that was adjusted for calendar period and birth cohort effects. The incidence rate crossover between black and white ethnic groups was robust (ie, reliable and reproducible) to these adjustments.

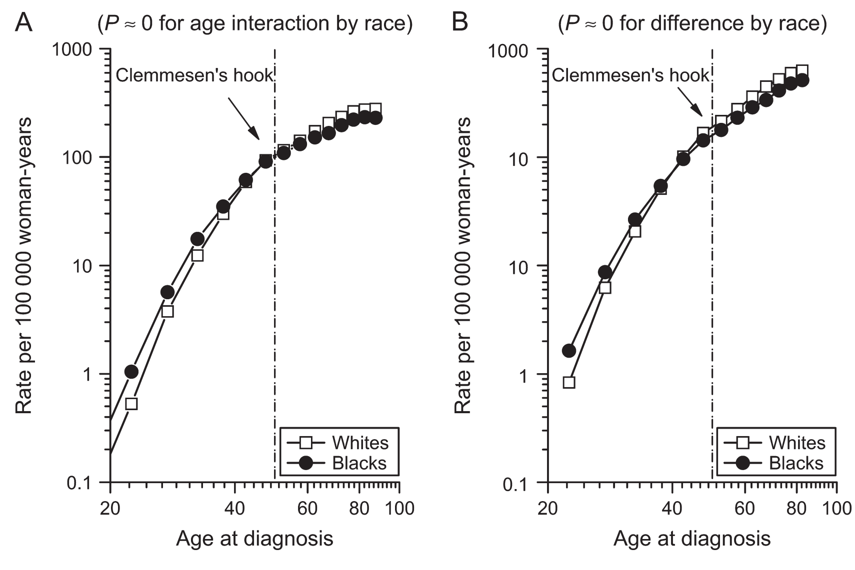

Table 1. Descriptive statistics of invasive breast cancer among women in the National Cancer Institute's SEER 9 Registries database from 1975 through 2004*

Variable	Total (n = 440 653) No.	Total Rate (95% CI)	Black women (n = 34 478) No.	Black Rate (95% CI)	White women (n = 381 122) No.	White Rate (95% CI)	IRR_{BW} (95% CI)
Age, y (1975+)							
<40	27 931	13.2 (13.1 to 13.4)	3961	15.5 (15.0 to 16.0)	21 638	13.1 (12.9 to 13.3)	1.2 (1.1 to 1.2)
40–49	72 066	152.0 (150.9 to 153.2)	7399	146.2 (142.9 to 149.6)	59 028	155.2 (154.0 to 156.5)	0.9 (0.9 to 1.0)
50–69	194 720	293.1 (291.8 to 294.4)	14 981	256.6 (252.5 to 260.7)	167 954	304.3 (302.8 to 305.7)	0.8 (0.8 to 0.9)
≥70	145 936	429.1 (426.9 to 431.3)	8137	361.3 (353.5 to 369.3)	132 502	446.1 (443.7 to 448.5)	0.8 (0.8 to 0.8)
Tumor size (1988+)							
≤2.0 cm	165 413	74.8 (74.1 to 75.1)	10 393	51.5 (50.5 to 52.5)	144 082	79.4 (79.0 to 79.9)	0.6 (0.6 to 0.7)
>2.0 cm	91 557	41.2 (41.0 to 41.5)	10 088	48.6 (47.6 to 49.5)	75 060	41.3 (41.0 to 41.6)	1.2 (1.2 to 1.2)
Other or unknown	183 683	—	13 997	—	161 980	—	—
Lymph node status (1988+)							
Negative	155 537	70.7 (70.3 to 71.0)	10 363	50.5 (49.6 to 51.5)	134 216	74.7 (74.3 to 75.1)	0.7 (0.7 to 0.7)
Positive	78 891	36.0 (35.8 to 36.3)	7614	35.9 (35.1 to 36.8)	65 881	37.1 (36.8 to 37.4)	1.0 (0.9 to 1.0)
Other or unknown	206 225	—	16 501	—	181 025	—	—
Tumor grade (1975+)							
Low	149 130	42.1 (41.9 to 42.3)	9017	29.9 (29.3 to 30.6)	129 690	43.6 (43.4 to 43.9)	0.7 (0.7 to 0.7)
High	109 642	31.3 (31.1 to 31.5)	11 621	36.6 (36.0 to 37.3)	91 169	31.3 (31.1 to 31.5)	1.2 (1.1 to 1.2)
Other or unknown	181 881	—	13 840	—	160 263	—	—
ER status (1990+)							
Positive	166 097	74.9 (74.6 to 75.3)	10 303	51.3 (50.3 to 52.3)	143 783	79.1 (78.7 to 79.5)	0.6 (0.6 to 0.7)
Negative	49 044	22.4 (22.2 to 22.6)	6609	30.9 (30.2 to 31.7)	38 784	21.9 (21.7 to 22.2)	1.4 (1.4 to 1.4)
Other or unknown	225 512	—	17 566	—	198 555	—	—

* Data for rate are the number of patients per 100 000 woman-years (age adjusted to the US standard population for the year 2000). Year in parenthesis indicates the time period in which SEER collected data for a given tumor characteristic. For example, SEER recorded age at diagnosis from 1975 forward (1975+). Because SEER did not collect and/or record all tumor characteristics for all time periods, rates and rate ratios are calculated only for those patients with breast cancer who had available data. SEER = Surveillance, Epidemiology, and End Results; CI = confidence interval; IRR_{BW} = black vs white incidence rate ratio in which the incidence rate among black women is compared with that among white women; ER = estrogen receptor; — = not calculated and/or not applicable.

cohort deviations that might differ between black and white women. For example, period effects could differ between black and white women if screening trends varied by race, and cohort effects could differ if the profile of breast cancer risk factors such as obesity or hormone replacement therapy changed differentially in successive cohorts of black and white women.

Kernel density estimation was used to produce smoothed age distribution curves at diagnosis (or density plots) in single years, as previously described (24). In brief, the kernel smoother estimated the underlying probability density function for breast cancer incidence by age at diagnosis in single years. The area under each density plot represented 100% of the patients with breast cancer; 95% confidence intervals were calculated with bootstrap resampling techniques.

Finally, two-component mixture models were used to determine whether the age distributions at diagnosis fit two or more breast cancer populations better than a single breast cancer population. The key model parameter was the probability of being in the earlier group, which was used to summarize changes in age distributions over time. The probability density function, g, is given by the formula: $g(y) = f(y;\alpha_0)p + f(y;\alpha_1)(1 - p)$, where $y = (x\lambda - 1)/\lambda$ and x = age at diagnosis with parameters (λ, p, α_0, α_1). The parameter λ denotes the power transformation, ρ is the mixing proportion, α_0 and α_1 stand for all the parameters that are used in the component densities f. Further details on the parameterization and estimations have been described (9,10). We used the log-likelihood ratio test and the Akaike information criterion to determine model fit, with the larger Akaike information criterion corresponding to the model with the better fit to the data (13). All statistical tests were two-sided.

Results

Descriptive Statistics

The SEER 9 Registries database collected data on 440 653 patients with invasive female breast cancer who were newly diagnosed during the period from January 1, 1975, through December 31, 2004 (Table 1). Among these 440 653 patients, 34 478 were black and 381 122 white. Black patients had a younger mean age at diagnosis (57.6 years) than white patients (62.6 years) and a larger mean tumor size (2.8 vs 2.1 cm) (for both, $P < .001$ for difference). Overall breast cancer incidence was lower among black women (111.9 cases of breast cancer per 100 000 woman-years) than among white women (128.5 cases of breast cancer per 100 000 woman-years) (difference = 16.6 cases of breast cancer per 100 000 woman-years, 95% confidence interval [CI] = 16.5 to 16.7). However, among women younger than 40 years (Table 1), incidence rates were 20% higher for black women than for white women (IRR_{BW} = 1.2, 95% CI = 1.1 to 1.2), whereas among women aged 40 years or older, incidence rates were lower for black women than for white women (IRR_{BW} < 1.0 for age groups 40–49, 50–69, and ≥70 years). Of note, the incidence rate ratio for black to white women was less than 1.0 for all low-risk tumor characteristics (eg, IRR_{BW} = 0.6, 95% CI = 0.6 to 0.7, for tumor size ≤2.0 cm) and 1.0 or more for all high-risk tumor characteristics (eg, IRR_{BW} = 1.2, 95% CI = 1.2 to 1.2, for tumor size >2.0 cm) (25). Similar patterns

Table 2. Temporal trends in invasive breast cancer among women in the National Cancer Institute's SEER 9 Registries database for three time periods: 1975 through 1979, 1990 through 1994, and 2000 through 2004*

	1975–1979 (n = 48 285)		1990–1994 (n = 79 642)		2000–2004 (n = 94 446)		
Variable	**No.**	**Rate (95% CI)**	**No.**	**Rate (95% CI)**	**No.**	**Rate (95% CI)**	**%CH (95% CI)**
Age, y (1975+)							
<40	3263	12.6 (12.1 to 13.0)	5099	13.0 (12.6 to 13.3)	5109	13.3 (12.9 to 13.7)	5.8 (5.6 to 6.0)
40–49	7792	138.3 (135.3 to 141.5)	13 429	159.4 (156.4 to 162.1)	16 602	152.7 (150.4 to 155.1)	10.4 (10.1 to 10.7)
50–69	23 344	233.0 (230.0 to 236.0)	33 098	308.2 (304.9 to 311.6)	42 239	321.7 (318.7 to 324.8)	38.1 (35.3 to 41.1)
≥70	13 886	333.8 (339.4 to 462.7)	28 016	462.7 (457.3 to 468.2)	30 496	441.0 (436.1 to 446.1)	32.1 (29.1 to 35.5)
Race (1975+)							
White	43 592	104.8 (103.8 to 105.8)	68 838	135.8 (134.8 to 136.8)	78 566	136.5 (135.5 to 137.4)	30.2 (29.3 to 31.1)
Black	3123	87.8 (84.6 to 91.1)	6357	120.0 (117.0 to 123.1)	8349	117.8 (115.3 to 120.4)	34.2 (30.3 to 38.5)
Other or unknown	1570	—	4447	—	7531	—	—
Tumor size (1988+)							
<2.0 cm	—	—	43 498	71.8 (71.1 to 72.4)	55 342	77.1 (76.5 to 77.8)	7.5 (7.4 to 7.5)
≥2.0 cm	—	—	24 580	40.6 (40.1 to 41.1)	29 508	40.8 (40.3 to 41.2)	0.5 (0.5 to 0.5)
Other or unknown	—	—	11 564	—	9596	—	—
Lymph node status (1988+)							
Negative	—	—	41 927	69.6 (68.9 to 70.2)	51 719	72.5 (71.8 to 73.1)	4.2 (4.2 to 4.3)
Positive	—	—	21 115	35.3 (34.8 to 35.8)	27 339	38.2 (37.8 to 38.7)	8.2 (8.1 to 8.3)
Other or unknown	—	—	16 600	—	15 388	—	—
Tumor grade (1975+)							
Low	3013	6.4 (6.2 to 6.6)	27 723	45.7 (45.2 to 46.2)	53 817	74.7 (74.1 to 75.3)	1066.6 (400.3 to 2842.0)
High	5471	11.5 (11.8 to 38.0)	22 818	38.0 (37.5 to 38.5)	31 184	43.5 (43.0 to 44.0)	277.4 (214.6 to 358.5)
Other or unknown	39 801	—	29 101	—	9445	—	—
ER status (1990+)							
Positive	—	—	45 254	74.4 (73.7 to 75.1)	64 126	89.1 (88.4 to 89.8)	19.7 (19.3 to 20.0)
Negative	—	—	14 604	24.6 (24.2 to 25.0)	17 516	24.5 (24.1 to 24.9)	−0.5 (−0.5 to −0.6)
Other or unknown	—	—	19 784	—	12 804	—	—

* Data for rate are the number of patients per 100 000 woman-years (age adjusted to the US standard population for the year 2000). Year in parenthesis indicates the time period in which SEER collected data for a given tumor characteristic. For example, SEER recorded age at diagnosis from 1975 forward (1975+). Because SEER did not collect and/or record all tumor characteristics for all time periods, rates and percent change are calculated only for those patients with breast cancer who had available data. Rate = the number of patients diagnosed with breast cancer per 100 000 woman-years (age adjusted to the US standard population for the year 2000); SEER = Surveillance, Epidemiology, and End Results; CI = confidence interval; %CH = percent change from 1975–1979 or 1990–1994 to 2000–2004; ER = estrogen receptor; — = not calculated and/or data not available.

were observed for lymph node–negative vs lymph node–positive disease, low- vs high-grade tumor, and ER-positive vs ER-negative disease.

Temporal trends in incidence rates among women with invasive breast cancer are shown in Table 2 for the following three time periods: 1975–1979, 1990–1994, and 2000–2004. The median age at diagnosis rose from 61 to 64 years from 1975–1979 to 1990–1994, and then returned to 61 years during 2000–2004. Incidence rates overall increased 28.3% (95% CI = 27.5% to 29.0%) from 1975–1979 to 2000–2004 (ie, from 102.1 to 131.0 cases of breast cancer per 100 000 woman-years, respectively). Incidence rates increased more over this period for older women (for ages 50–69 years, 38.1%, 95% CI = 35.3% to 41.1%) than for younger women (for ages <40 years, 5.8%, 95% CI = 5.6% to 6.0%). Among both black and white women, incidence rates rose substantially from 1975–1979 to 1990–1994, with little or no increase from 1990–1994 to 2000–2004 (26). Incidence rates generally rose more for tumors with favorable characteristics (eg, small tumors, lymph node–negative disease, low-grade tumor, and ER-positive status) than for tumors with unfavorable characteristics (eg, large tumors, lymph node–positive disease, high-grade tumor, and ER-negative status), although conclusions are somewhat limited by the reapportionment of unknown to known tumor characteristics over time. For example, percent unknown for ER status fell from 24.8% (or 19 784 of a total of 79 642 patients) during 1990–1994 to 13.6% (or 12 804 of a total of 94 446 patients) during 2000–2004. Additionally, given that SEER did not record tumor characteristics for all time periods, time trends for tumor characteristics also were relatively short for tumor size (from 1988 forward), lymph node status (from 1988 forward), and hormone receptor expression (from 1990 forward). Therefore, these percent changes should be interpreted with caution.

Incidence Patterns by Age and Race

Age-adjusted incidence rates rose until the late 1990s and then began to fall (Figure 1, A), with rates greater for white women than for black women for the entire study period (1975–2004). Notwithstanding these overall secular trends, age-specific temporal trends varied "qualitatively" (ie, they switched, crossed, or reversed) by race and age (Figure 1, B). For example, among women aged 40–49, 50–69, and 70 years or older, incidence rates were greater for white women than for black women for the entire study period, whereas among women younger than 40 years, incidence rates were greater for black women. Of note, measures of interaction (effect modification) by race within age groups were statistically significantly different for women aged 40 years or older (ranging from $P < .01$ to $P < .001$) but not statistically significantly different among women younger than 40 years ($P = .75$).

Age-specific incidence rates also varied qualitatively (ie, they switched, crossed, or reversed) by race (Figure 2). For example, overall age-specific incidence rates rose rapidly until age 40–50 years for both white and black women and then continued to rise at a slower pace (Figure 2, A). This midlife change, pause, or inflection in age-specific incidence rates near age 50 years has been termed Clemmesen's hook and has been attributed to menopause (27). Notwithstanding overall age-specific incidence rates, age-specific incidence rates were greater among black women than among white women who were younger than 40 years, and then, incidence rates were higher for white women ($P < .001$ for age × race interaction). The age–period–cohort fitted age-at-onset curves (Figure 2, B; adjusted for calendar period and birth cohort effects) also were higher among black women who were younger than 40 years than among white women in the same age group and higher among white women who were aged 40 years or older than among black women in the same age group ($P < .001$ for difference by race). Additionally, age–period–cohort analyses detected statistically significantly different age deviations among black and white women near the age of 40 years ($P < .001$ for difference by race), differential calendar period effects during the 1980s ($P < .001$ for difference by race), and differential birth cohort effects near the 1950 birth cohort ($P = .009$ for difference by race).

The black-to-white incidence rate crossover was observed for all tumor characteristics (Figure 3), although at earlier ages at diagnosis for low-risk tumor characteristics than for high-risk tumor characteristics. For example, the crossing of black-to-white incidence rates occurred at ages 25–29 years for tumor sizes 2 cm or less, whereas there was no crossover for tumor sizes more than 2 cm. For lymph node–negative and lymph node–positive disease, incidence rates crossed at ages 35–39 and 50–54 years, respectively. For low- and high-grade tumors, incidence rates crossed at ages 30–34 and 80–84 years, respectively. For ER-positive tumors, the crossing of incidence rates for black and white women occurred at ages 30–34 years, but there was no crossover for ER-negative tumors (28).

The shapes of the black and white age-specific incidence rate curves (Figure 2) reflected bimodal early- and late-onset breast cancer populations, with modes (ie, peak frequencies) near ages 50 and 70 years, respectively (Figure 4). These bimodal breast cancer

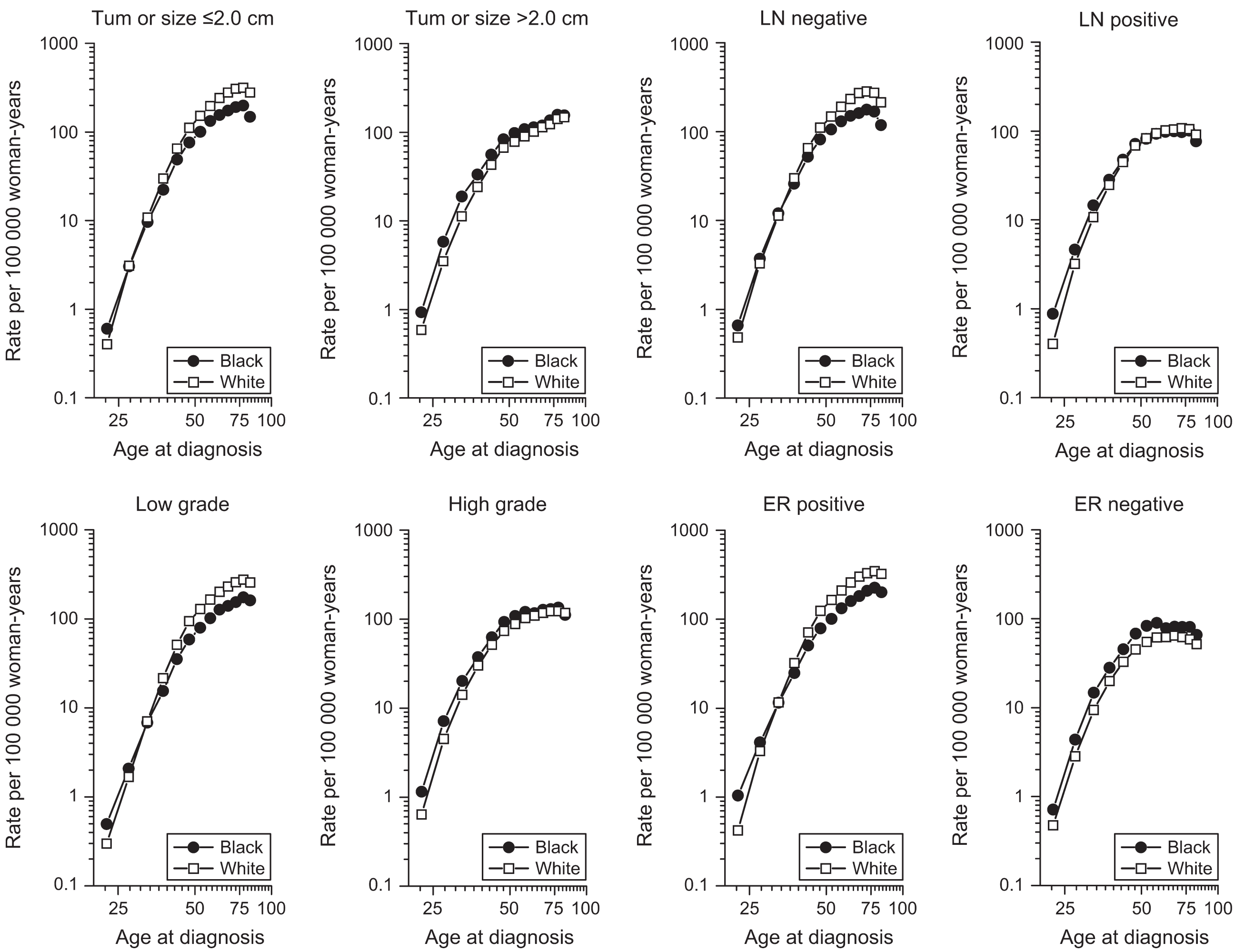

Figure 3. Incidence rate crossover between black and white women stratified by tumor characteristics (tumor size, LN status, tumor grade, and ER status) in the National Cancer Institute's Surveillance, Epidemiology, and End Results 9 Registries database from 1975 through 2004. The crossover was observed for all tumor characteristics (as indicated at the top of each panel), although more so for those with low-risk disease (tumor size ≤2.0 cm, axillary LN-negative disease, low-grade tumor, or ER-positive status) than for those with high-risk disease (tumor size >2.0 cm, axillary LN-positive disease, high-grade tumor, or ER-negative status). LN = lymph node; ER = estrogen receptor.

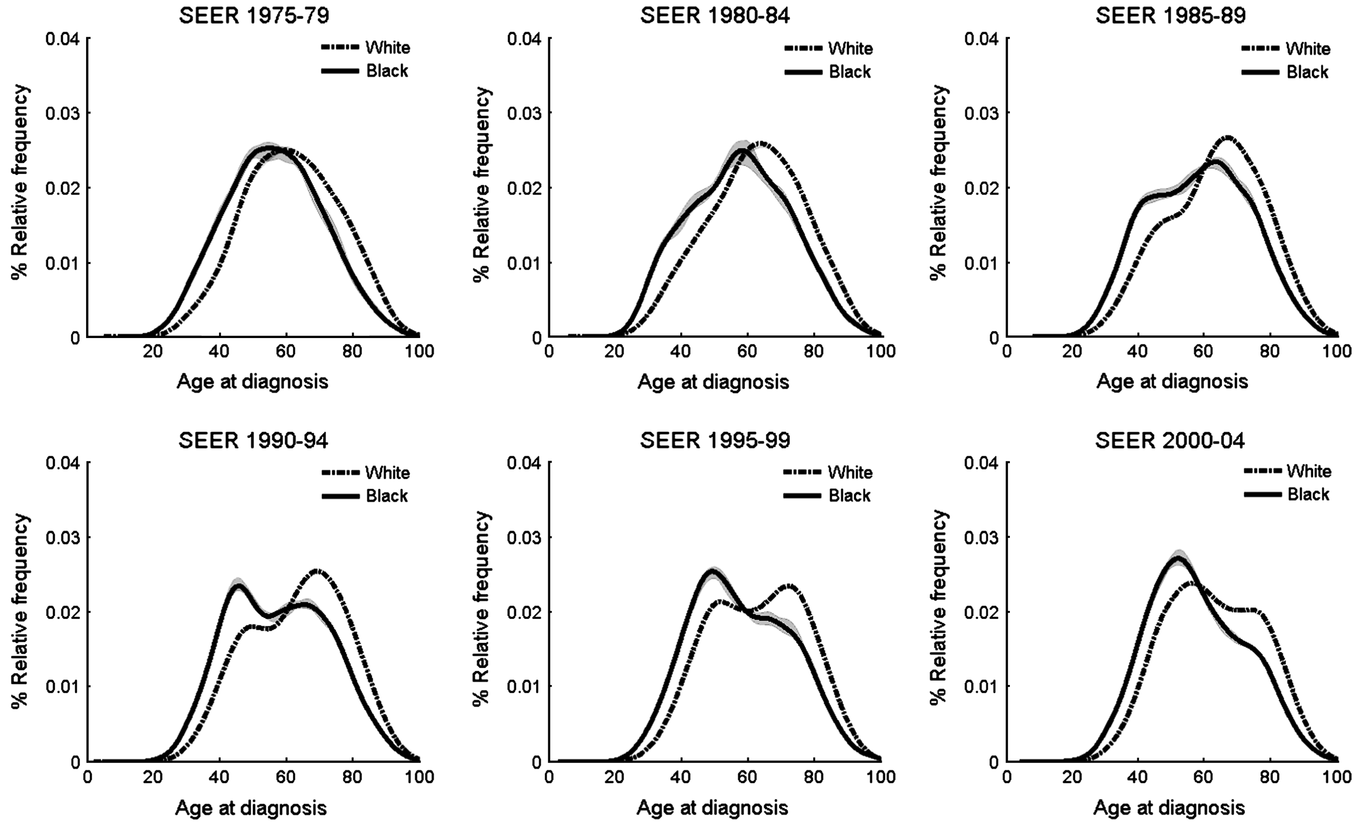

Figure 4. Age distributions at diagnosis by race (black vs white) and time period (1975–1979, 1980–1984, 1985–1989, 1990–1994, 1995–1999, and 2000–2004) in the National Cancer Institute's SEER 9 Registries database from 1975 through 2004. Bimodal age distributions are expressed as percent relative frequency among white and black women; 95% confidence intervals are shown as **shaded bands** and were calculated by using bootstrap resampling techniques. Of note, 95% confidence intervals were too narrow to be discernable for white women and are only slightly visible for black women, providing further evidence for distinct bimodal breast cancer populations among white and black women. Additionally, bimodal breast cancer populations fluctuated over time among white and black women. SEER = Surveillance, Epidemiology, and End Results.

populations fluctuated over time. From 1975–1979 to 1985–1989, breast cancer populations shifted to predominant late-onset age distributions for black and white women. Beginning in 1990–1994, population distributions of black and white breast cancer populations began to return to dominant early-onset age distributions. By 2000 through 2004, early-onset breast cancer populations were reestablished for both black and white women. Of note, the movement of black women toward later ages at onset during the 1980s was slower than that of white women, but it was faster to return toward earlier ages at onset in the 1990s.

Two-component mixture models confirmed that two or more breast cancer populations fit the data better than a single cancer population for both black and white women (Table 3). For example, during 1975–1979, Akaike information criterion among white women was larger for a mixture than for a single cancer population. We observed similar patterns for all time periods and for both races. Additionally, the probability for being in the early-onset breast cancer population decreased in the 1980s and then increased in the late 1990s (Table 3; Figure 5), consistent with the density plots in Figure 4. For example, among white women, the probability for early age at onset fell from 77% (95% CI = 73.0% to 80.9%) in 1975 through 1979 to 25% (95% CI = 23.0% to 27.0%) in 1985–1989, and then rose from 33% (95% CI = 31.8% to 34.2%) in 1990–1994 to 64% (95% CI = 62.2% to 65.8%) in 2000–2004. Black women had a similar pattern.

Incidence Patterns by Age, Race, and ER Status

Age-specific incidence rates were higher for ER-negative tumors among younger women and higher for ER-positive tumors among older women (Supplementary Figure 1, A, available online). Poisson regression models confirmed the interaction between age and ER expression ($P < .001$ for age × ER interaction). ER Furthermore, for all age groups and all time periods, incidence rates for ER-positive tumors were greater among white women than among black women (Supplementary Figure 1, B, available online). In contrast, for all age groups and all time periods, incidence rates for ER-negative tumors were greater among black women than among white women (Supplementary Figure 1, C, available online). For both ER-positive and ER-negative tumors, measures of interaction between age groups and race were statistically significantly different for older women ($P <$ .001 among black and white women with ER-positive tumors and aged ≥70 years, and $P < .001$ among black and white women with ER-negative tumors and aged ≥70 years) but not for younger women ($P = .67$ among black and white women with ER-positive tumors and aged <40 years, and $P = .16$ among black and white women with ER-negative tumors and aged <40 years).

Table 3. Estimates for mean ages at diagnosis of patients with early- or late-onset breast cancer from two-component mixture models*

Calendar period	Race	Mean age, y (95% CI)		Probability†, % (95% CI)	AIC	
		Early-onset group	Late-onset group		Mixture	Single density
1975–1979	White	47.9 (34.6 to 61.2)	83.4 (70.1 to 96.7)	77.0 (73.0 to 80.9)	−177052	−177060
	Black	42.4 (−41.9 to 126.7)	69.8 (−14.5 to 154.1)	43.0 (15.6 to 70.4)	−12706	−12716
1980–1984	White	52.1 (48.2 to 56.0)	79.0 (75.1 to 82.9)	69.0 (65.1 to 72.9)	−203304	−203314
	Black	32.0 (13.2 to 50.8)	51.6 (32.8 to 70.4)	17.0 (11.1 to 22.9)	−16180	−16188
1985–1989	White	39.2 (31.2 to 47.2)	67.9 (59.9 to 75.9)	25.0 (23.0 to 27.0)	−257709	−257717
	Black	36.3 (7.1 to 65.5)	61.4 (32.2 to 90.6)	15.9 (5.3 to 26.5)	−21259	−21269
1990–1994	White	46.8 (46.6 to 47.0)	71.0 (70.8 to 71.2)	33.0 (31.8 to 34.2)	−280278	−280286
	Black	39.1 (26.0 to 52.2)	60.4 (47.3 to 73.5)	20.0 (10.2 to 29.8)	−26053	−26063
1995–1999	White	46.2 (43.8 to 48.6)	73.5 (71.1 to 75.6)	48.0 (46.4 to 49.6)	−312921	−312929
	Black	46.7 (43.3 to 50.0)	70.9 (67.6 to 74.2)	56.3 (51.4 to 61.2)	−30803	−30808
2000–2004	White	46.8 (45.6 to 48.0)	76.9 (75.7 to 78.1)	64.0 (62.2 to 65.8)	−319002	−319010
	Black	50.2 (47.5 to 52.9)	73.7 (71.0 to 76.4)	68.7 (64.2 to 73.2)	−33939	−33945

* AIC were used to determine model fit, with the larger values corresponding to the model with the best fit to the data. For white and black women, a mixture model (ie, more than one breast cancer population) fit the data better than a single cancer population (single density model). CI = confidence interval; AIC = Akaike information criteria.

† Probability of being in the early-age-at-onset group at the time of initial breast cancer diagnosis.

Age and trend interactions by ER status (Supplementary Figure 1, available online) reflected bimodal early- and late-onset breast cancer populations (Supplementary Figure 2, available online). Beginning in the 1990s, the bimodal population of ER-positive tumors began to shift from being dominant at later ages to earlier ages at onset. The shift toward earlier ages at onset was exaggerated for ER-negative breast cancers.

Discussion

The crossover in incidence rates between black and white ethnic groups was a robust (ie, reliable and reproducible) finding in the SEER 9 Registries database for the entire study period of 1975 through 2004 (Figure 1). This age-related interaction (or effect modification) was statistically significant in Poisson regression models, and it remained so in age–period–cohort models, which were adjusted for calendar period and birth cohort effects (Figure 2). In women younger than 40 years, incidence rates were statistically significantly higher among black women than among white women, whereas the reverse was true among women aged 40 years or older for all time periods. The incidence rate crossover between black and white ethnic groups was also observed after stratification of women by tumor characteristics, although the crossover occurred at earlier ages for low-risk prognostic characteristics than for high-risk prognostic characteristics (Figure 3). Furthermore, age distributions at diagnosis shifted over time, as illustrated with density plots (Figure 4) and mixture models (Table 3; Figure 5), which tracked a secular rise and fall in the proportion of late-onset cases of breast cancer among white women and a similar but delayed pattern among black women.

When recognized as a qualitative (crossing or reversing) age interaction, the breast cancer incidence rate crossover between black and white ethnic groups can be seen as a manifestation of breast cancer heterogeneity, with black women having more of the early-onset breast cancer types and less of the late-onset breast cancer types than white women. Given the early-onset predominance for ER-negative breast cancers (Supplementary Figure 1, A, available online), it also is not surprising for ER-negative tumors to be more common among black women than among white women. It should be noted that although black and white women have different proportions of early- and late-onset breast cancer types, these results do not indicate that the two racial groups have different kinds of breast cancer, although black women have proportionally more of the worse types (ie, early-onset and ER-negative tumors).

Throughout this study, we have focused on the qualitative age interactions among black and white women, but similar crossing, reversing, or dual age-dependent effects have also been observed for breast cancer risk factors, such as parity and obesity (29–35). That is, for younger women, parity is associated with increased risk and obesity is associated with decreased risk, whereas for older women, parity is associated with decreased risk and obesity is associated with increased risk. Therefore, differential racial profiles for risk factors with qualitative age-related effects (eg, parity and obesity) might account for at least part of the incidence rate crossover between black and white ethnic groups (3,5–7). However, age–period–cohort models confirmed an age-specific black-to-white incidence rate crossover, even after adjustment for birth cohort (risk factor) effects. Thus, the black-to-white ethnic crossover cannot be fully explained by known risk factor variations (36).

Additionally, the black-to-white incidence rate crossover has been present despite well-documented period and cohort racial variations over time. For example, breast cancer incidence rates more than doubled from 1935 to 1998 before falling after the Women's Health Initiative report in 2002 (26,37–43), especially among older women with tumors of low malignant potential (44–46). The shift of black women toward older ages at diagnosis was slower than that of white women during the 1980s, and it was faster to return toward early ages at onset in the 1990s (Figure 4), possibly due to racial variations for calendar period (screening) and/or birth cohort (exposure) effects. That is, screening mammography initially lagged for black women but equalized by the mid-1990s (42,47), perhaps accounting for a slower initial rise in screen-derived late-onset breast cancers among black women.

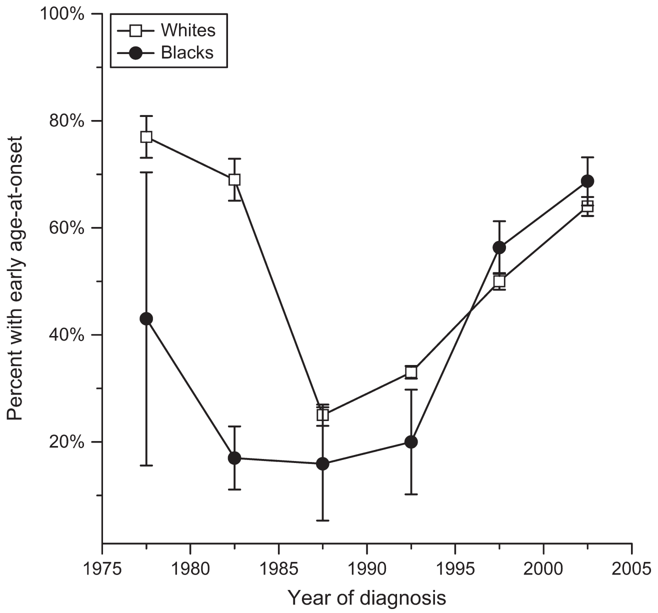

Figure 5. Percentage of breast cancer patients with an early age at onset by race in the National Cancer Institute's Surveillance, Epidemiology, and End Results 9 Registries database from 1975 through 2004. Error bars = 95% confidence intervals.

Additionally, although the percent changes in exposure to hormone replacement therapy before and after reports from the Women's Health Initiative were similar among racial groups (48), white women were approximately twice as likely as black women to use hormone replacement therapy (49), also potentially explaining a slower initial rise in postmenopausal breast cancers. Yet, age–period–cohort models confirmed an age-specific black-to-white incidence rate crossover, even after adjustment for these calendar period (screening) and birth cohort (exposure) effects.

Our study has the usual limitations of descriptive epidemiology (ie, retrospective registry assessment, missing data, nonstandardized histopathologic typing, and lack of individual-level risk factor data). A major strength was the supplementation of standard descriptive epidemiology with mathematical models. Therefore, although this ecologic study cannot determine the factors responsible for the incidence rate crossover, as far as we know, it is the first validation of this phenomenon in vital rates by use of a structured quantitative approach.

In summary, breast cancer–age interactions by black and white race suggest biologic heterogeneity of the qualitative variety (ie, they switched, crossed, or reversed). Although the black-to-white ethnic crossover has been attributed to a one-time (early) adverse effect and long-term risk reduction for reproductive risk factors such as parity (5), we suggest that the crossover reflects qualitative age interactions (effect modification), whereby risk factors such as parity have different effects on different types of breast cancer. Indeed, recent studies show differential racial and risk factor profiles for different breast cancer phenotypes (50–54). Empirical studies should attempt to verify this ecologic phenomenon with individual risk factor data. Future analytic studies also should carefully and probably routinely be powered to assess the interaction between age and race because if the incidence rate crossover from black to white ethnic groups is a proxy for an age-specific biologic heterogeneity, then early- and late-onset types of breast cancer are distinctly different diseases. Ignoring such heterogeneity may obscure important stratum-specific effects that result from subgroup averaging of nonpoolable and/or dissimilar breast cancer populations (14,55).

Appendix 1: Age–Period–Cohort Analysis

As discussed by Rosenberg and Anderson (unpublished material), let ρ_{ij} denote the logarithm of the expected incidence rates for age group j and calendar period i, which are defined by using equally spaced age and period intervals. In age–period–cohort analysis, we assume log–linear effects for age (α_j), period (π_i), and cohort (γ_k) over $j = 1, \ldots, J$ age groups, $i = 1, \ldots, I$ calendar periods, and $k = 1, \ldots, I + J - 1$ birth cohorts:

$$\rho_{ij} = \alpha_j + \pi_i + \gamma_k.$$

It is well recognized that this model is not identifiable; that is, the linear trends in age, period, and cohort effects, α_L, π_L, and γ_L, respectively, cannot be separated because of the collinear relation of the cohort, period, and age indices. That is, the year of birth equals the calendar period or year of diagnosis minus the age or age at diagnosis, or $k = i - j + J$, where i, j, and k index the observed age, period, and cohort effects. However, as described previously by Holford (19), it can be shown with the theory of estimable functions that the parameters in the restricted model with $\pi_L = 0$ yield fitted log–linear rates equal to:

$$\rho_{ij} = \mu + (\alpha_L + \pi_L)(j - \bar{j}) + (\pi_L + \gamma_L)(k - \bar{k}) + \tilde{\alpha}_j + \tilde{\pi}_i + \tilde{\gamma}_k.$$

Each of the terms in Equation 1 is identifiable, and the fitted rates are the same as all other solutions that maximize the likelihood. In this equation, $\bar{j}$ and $\bar{k}$ are the central or referent indices of age and cohort, respectively; the coefficient of the age trend $(j - \bar{j})$ provides an estimate of $(\alpha_L + \pi_L)$, and the coefficient of the cohort trend $(k - \bar{k})$ provides an estimate of $(\pi_L + \gamma_L)$. We designate $(\alpha_L + \pi_L)$ as the "longitudinal age trend" and $(\pi_L + \gamma_L)$ as the "net drift" (16,17,23,56). The parameter μ is the intercept term, and the parameters $\tilde{\alpha}_j$, $\tilde{\pi}_i$, and $\tilde{\gamma}_k$ are the orthogonal (and identifiable) deviations of Holford (19).

We also considered the function $\tilde{\rho}_j = \mu + (\alpha_L + \pi_L)(j - \bar{j}) + \tilde{\alpha}_j$, which we term the "fitted age-at-onset curve." This function provides a summary estimate of the longitudinal age-at-onset curve, which represents an extrapolation of the experience of the midcohort ($k = \bar{k}$ and assume that $\tilde{\gamma}_{\bar{k}} \approx 0$) that is based on the experience of all other cohorts included in the analysis. By construction, the fitted age-at-onset curve is adjusted for both period and cohort effects.

We fitted Equation 1 separately to incidence data for black and white women and used Wald tests to compare age–period–cohort parameters by race for the fitted age-at-onset curves; drift parameters; and age, period, and cohort deviations. For example, the Wald test statistic that was used to compare the net drifts has a chi-square distribution with 1 *df*:

$$\chi_1^2 = \frac{\left[(\pi_L + \gamma_L)^1 - (\pi_L + \gamma_L)^0\right]^2}{(SD^1)^2 + (SD^0)^2},$$

where SD^1 and SD^0 are the estimated standard deviations of the net drift parameter $(\pi_L + \gamma_L)$ for black and white women, respectively. Similarly, Wald tests were developed to test for parallel age-at-onset curves ($I - 1$ *df*) and for equal age deviations ($I - 2$ *df*), period deviations ($J - 2$ *df*), and cohort deviations ($I + J - 3$ *df*).

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